

Coconut

Andhra Pradesh is one of the states that grows coconut extensively. In our state Andhra Pradesh, coconut cultivation is being carried out on 1.17 lakh hectares. In terms of area, Andhra Pradesh state ranks 4th after Kerala, Tamil Nadu, and Karnataka, and is in the first position in terms of productivity. More than half of the area is in the twin Godavari districts, and the rest is in Guntur and Chittoor districts. Although our state is ahead in productivity, there is still a lot of scope to increase the yield in the North Coast and Krishna areas. If scientific and modern farming methods are followed, coconut farmers can increase their income along with the yield. Under the Dr. Y.S.R. Horticultural University, extensive research on coconut crops has been carried out for the past 70 years at the Horticultural Research Station, Ambajipeta. The appropriate scientific management practices to achieve good yield and quality in coconut are described below.

Climate and Soils

Coastal areas with high humidity and areas with high and regular rainfall are suitable. Annual rainfall should be between 1000 to 2000 mm. Fertile delta lands with irrigation and drainage facilities are very suitable for this crop. Gravelly and red soils with irrigation facilities are also suitable for cultivation. Inland areas far from the coastal region are not suitable for coconut cultivation. If the average temperature drops below 15 degrees Celsius, the growth of the coconut plant is significantly reduced.

Coconut Varieties

There are many varieties of coconut. The varieties mentioned below give high yields under the climatic conditions of our state.

1. **“East Coast Tall” (Desi):** This variety is widely grown in our “East Coast” regions. It usually comes to nut formation in 7 years and yields an average of 75 to 100 nuts per tree per year. The nut contains 146 grams of copra and 64 percent oil.
2. **“Godavari Ganga” (“HYBRID”):** Scientists at the Ambajipeta Coconut Research Station developed this "HYBRID" variety in 1991 using “East Coast Tall” (Desi) as the mother tree and Ganga Bondam as the male tree. This "HYBRID" was released for cultivation in our state under the name “Godavari Ganga”. It has also been recommended for cultivation in Andhra Pradesh and Tamil Nadu at the national level. This variety comes to nut formation in four years and starts giving good yields in 6 to 7 years. It yields an average of 140 nuts per tree per year. The nut contains 150 grams of copra with 68 percent oil.
3. **“Double Century”:** This variety yields more than the Desi variety and has been found suitable for cultivation in the “East Coast” region. This variety was released in 1994. It

comes to nut formation in about 7 years and yields an average of 130 nuts annually. The nuts of this variety contain 160 grams of copra and 64 percent oil.

4. **“Gauthami Ganga”:** This variety was selected from the “Ganga Bondam” dwarf variety and released for cultivation in the state in 2017. It is very useful for tender coconuts. It comes to nut formation within 4 years of planting. It yields 85 to 94 nuts per tree annually. The nut averages 157 grams of copra and 69 percent oil.
5. **“Vasishtha Ganga”:** This "HYBRID" variety was developed by crossing the “Ganga Bondam” dwarf variety and the “Philippines Ordinary” Tall variety and released for cultivation in Andhra Pradesh and Karnataka states at the national level in 2014. It comes to nut formation within 4 years of planting. It yields an average of 125 nuts per tree per year. The nut contains 158 grams of copra and 69 percent oil.
6. **“Vainateya Ganga”:** This "HYBRID" variety was developed by crossing the “Philippines Ordinary” Tall variety and the “Ganga Bondam” dwarf variety and released for cultivation in Andhra Pradesh in 2017. This "HYBRID" variety comes to flower within 4 years of planting. It yields an average of 118 nuts per tree per year. The nut contains 190.5 grams of copra and 66 percent oil.
7. **“Abhaya Ganga”:** This "HYBRID" variety was developed by crossing the “Ganga Bondam” dwarf variety and the “Laccadive Ordinary” Tall variety. It was released in 2017. This variety comes to flower within 4 years of planting and yields an average of 136 nuts per tree per year. The nut contains 170 grams of copra and 72 percent oil.

Characteristics of the tall, dwarf and "HYBRID" varieties:

a) Dwarf Varieties:

1. **Size:** Small, typically 5 to 12 meters (15 to 40 feet) tall.
2. **Bearing Age:** Very early, start nut formation within 2 to 5 years of planting.
3. **Lifespan:** Shorter economic life, usually 30 to 60 years.
4. **Pollination:** Mostly self-pollinated, leading to uniform offspring.
5. **Adaptability:** Generally less hardy, can be more susceptible to drought and certain diseases.
6. **Nut Characteristics:** Produce smaller nuts, often harvested for tender coconut water due to their sweet taste. Lower copra and oil content.
7. **Planting Density:** Higher density planting is possible due to their smaller canopy.
8. **Primary Use:** Popular for tender coconut water, ornamental purposes, and home gardens.
9. **Identification:** Smooth, pale stem with small gaps (under 2.5 cm) between leaf scars.

b) Tall Varieties:

1. **Size:** Large, growing to impressive heights of 15 to 30 meters (50 to 100 feet) or more.
2. **Bearing Age:** Later to mature, typically starting to nut formation after 6 to 10 years.
3. **Lifespan:** Longer economic life, often producing for 60 to 100 years.
4. **Pollination:** Predominantly cross-pollinated, leading to greater genetic variation.
5. **Adaptability:** Hardier and more adaptable to a wider range of soil types and environmental conditions. Generally more resistant to pests and diseases.
6. **Nut Characteristics:** Produce larger nuts with higher copra and oil content, making them ideal for commercial processing.
7. **Planting Density:** Lower density planting due to their large fronds and extensive canopy.
8. **Primary Use:** Mainstay for large-scale commercial plantations for copra and oil production.
9. **Identification:** Rougher stem with larger gaps (over 5 cm) between leaf scars.

c) "HYBRID" Varieties:

1. **Description:** These are created by cross-breeding dwarf and tall varieties (e.g., Dwarf x Tall or Tall x Dwarf).
2. **Size:** Medium height, typically ranging from 10 to 20 meters.
3. **Bearing Age:** Early to medium, often starting to nut formation within 4 to 7 years.
4. **Lifespan:** Medium economic life, generally 50 to 80 years.
5. **Pollination:** Cross-pollinated, as they are a result of controlled crosses.
6. **Adaptability:** Generally offer a good balance of adaptability and often show improved disease tolerance compared to pure dwarfs.
7. **Nut Characteristics:** Aim to combine the best of both parents. usually produce a good number of nuts with higher copra and oil content, while also providing good quality tender coconut water.
8. **Planting Density:** Medium density planting.
9. **Primary Use:** Highly favored for commercial plantations due to their excellent yield, early bearing, and combined desirable traits from both parent types.
10. **Identification:** Exhibit a mix of characteristics from their dwarf and tall parents, often showing vigorous growth.

In essence, "HYBRID" coconut varieties are bred to combine the best attributes of native tall and dwarf varieties, resulting in plants that are more productive, bear earlier, and often have improved disease resistance, making them highly desirable for commercial cultivation. Native varieties, while offering different advantages (like longevity in tall types or early sweetness in dwarfs), may not always match the overall productivity and efficiency of "HYBRID" under optimal management.

Farmers queries:

- 1. I am planning to establish a new coconut garden in my village. Should I buy big plants of 3 to 4 years and plant them or should I buy 1 year old plants?**

It is always suggested to plant small plants of 12 to 18 months to ensure better root establishment and adaptability to the climate. Planting bigger trees of 3 to 4 years draws problems like poor establishment as they grow in restricted environments inside the bags and these are prone to fall off during heavy winds and cyclones. The adaptability is also less compared to the small plants.

- 2. I am planning to establish a new coconut garden in my village. Should I buy native variety plants or "HYBRID" varieties?**

It is recommended to go for native varieties of coconut instead of "HYBRID"s. If looking for a more disease resistant and long term crop. If we need early crop in 4 years then "HYBRID"s must be a better option but utmost care may be taken in fertilisers and maintenance so as to ensure high yields. Studies observed that native varieties maintained good field practices on par with "HYBRID" and gave good results .

Cultivation Methods

Seed Nut Selection and Nursery Preparation Method in Coconut

Seed coconuts should be collected only from selected mother trees with good characteristics. The mother trees in the selected orchard for seed should have high yield (average 100 nuts/year/tree) and nut quality (150 g copra per nut). The age of the trees in the orchard should be 15 to 40 years. Mother trees should be healthy and consistently high yielding. Only fully matured nuts (12 months old) on the tree during April to May should be collected for seed. Seed nuts should be dried in the shade for 20 days and planted in the nursery in June.

Nursery Preparation

Light upland soils with drainage facilities are suitable for nursery preparation. In heavy soils where water stagnates during the rainy season, a raised nursery should be prepared. Plant in the nursery with 30 cm between rows and 10 to 15 cm between two nuts in a row. Then, irrigate frequently to keep the moisture from drying out. Weeds should be controlled. Pests and diseases should be controlled with plant protection measures. Generally, the seed germinates within 2 months of planting. The germination rate is 65 to 70 percent.

Coconut Seedling Selection and Planting Method

To get a consistently good yield in coconut, quality seedlings must be selected and planted. Seedlings that germinate early in the nursery, have more leaves, a thick base (10 cm), and leaves that unfold quickly can be considered quality seedlings. Seedlings with these characteristics

should be selected. Seedlings should be healthy and free from any pests and diseases. It is a good practice to plant only 12 to 18 months old seedlings in the orchard. Generally, in summer, i.e., April to May, pits of size 1 meter x 1 meter x 1 meter should be dug at a distance of 8 meters x 8 meters and allowed to dry well. After the onset of monsoon in June to July, the pit should be filled with good soil, silt, and well-composted farmyard manure (25 kg) + 500 grams single super phosphate. Selected varieties should be planted at the rate of 60 seedlings per acre in the center of the pit during the early monsoon, filled with soil, the soil pressed firmly, and immediately irrigated lightly.

Recommended spacing for planting coconut gardens is 8 meters X 8 meters for all types of suitable soils across the andhra pradesh.

Water Management: Water should be supplied to coconut trees properly, considering the humidity, soil type, and climate. Delta regions flood irrigation is recommended. However, with this method, irrigation water is wasted, and diseases can spread quickly. It is better to provide water through the "basin method" by making a basin around the tree. Water can be conserved by providing water to the gardens through the drip method. In black soils, water must be provided once every 20 days, and in light red soils, once every 10 days. In light soils, it is necessary to water every 5 to 7 days in the summer months. Care should be taken in coconut gardens to avoid water stress. Due to water stress, the shedding of buds and a reduction in the size of the fruit occur.

Symptoms and Effects of Water Stress: In the summer months, an average of 50 to 60 liters of irrigation water is required per day for a bearing tree (4 to 60 years coconut plant). Farmers are unable to provide water to coconut gardens as recommended, as the groundwater levels are falling. As a result, coconut gardens are experiencing water stress. Due to this water stress, symptoms like the leaves of mature trees withering and hanging down, fewer flowers, and increased bud drop can be seen in the gardens. New leaves also grow less due to water stress. In the summer months, when the day temperatures are high and hot winds blow, flower fertilization decreases, and even bud formation decreases significantly. Moreover, coconut yield, fruit size, and coconut quality also decrease. Therefore, in the summer months, it is necessary to keep moisture by spreading coconut husk or coconut leaves or coconut husk in the plate of the tree.

Irrigation Method:

Drip Irrigation vs. Flood Irrigation

Drip irrigation is highly recommended and superior to flood irrigation for coconut cultivation.

Here's why:

1. Water Efficiency: Drip irrigation delivers water directly to the root zone, minimizing waste due to evaporation, runoff, and overspray. It can save 60 to 70% of water compared to traditional methods like flood irrigation.

2. Increased Yield and Profitability: Studies have shown that drip-irrigated coconut farms have higher net returns and better benefit-cost ratios compared to flood-irrigated farms, primarily due to increased yield and better production efficiency.

3. Reduced Weed Growth and Pest Incidence: By applying water only where needed, drip irrigation discourages weed growth, reducing labor for weeding.

4. Nutrient Efficiency: Fertilizers can be applied precisely through drip irrigation systems (fertigation), ensuring nutrients reach the plant roots directly and uniformly.

5. Conservation: It helps conserve water, a crucial resource, especially with increasing water scarcity.

While the initial setup cost for drip irrigation might be higher than flood irrigation, the long-term benefits in terms of water saving, increased yield, and reduced labor make it a more sustainable and profitable choice for coconut cultivation.

Management of Fertilizers: Coconut trees have flowering, fruit set, and developing fruits throughout the year. The coconut tree absorbs nutrients from the soil throughout the year. Therefore, it is important to ensure that nutrients are available in the soil for the coconut tree. Nitrogen, phosphorus, and potassium are the major nutrients for achieving high yields in coconut trees. Therefore, to achieve good yields and quality in coconut gardens, organic and chemical fertilizers should be provided every year.

Fertilizer Management Through Drip Method: When fertilizers are given to the coconut tree through the drip method, 75% of the recommended dosage, i.e., 375 grams of nitrogen (800 grams of urea) and 1125 grams of potash (muriate of potash 1870 grams), should be provided to the tree in eight installments from October to May. Phosphorus fertilizers 240 grams (1500 grams of super phosphate) should be applied in two installments, at the base of the plant.

Fertilizer Application Schedule per Coconut Tree above 5 years old are Urea 100 grams, DAP 65 grams, muriate of potash 234 grams every month from october to may.

Role of Major Nutrients in Coconut

1. Nitrogen: Nitrogen plays an important role in the growth of young plants and rapid spathe developemnt. Research has shown that applying nitrogen along with potash in the right proportions to fruit-bearing trees increases fruit yield by approximately 28%.

2. Phosphorus: Phosphorus helps in making the base of young coconut seedlings thick and strong and in the formation of more leaves. Therefore, when planting seedlings in the field, mixing phosphorus (250g) with well-composted farmyard manure and soil before planting the seedling increases the chance of the seedlings growing strong. When applied to nut bearing trees along with nitrogen and potash fertilizers, phosphorus helps the roots to absorb the nitrogen present in the soil.

3. Potash: Potassium is the most important macronutrient in coconut orchards. Due to this, the plants come to nut formation quickly. The number of spathes increases, fertilization in the bunches occurs properly, and there is a chance for the crop to stabilize. The yield of copra and oil in the nuts increases significantly. Due to potash, plants have the ability to withstand pests, diseases, and water stress.

Major Nutrient Deficiency Symptoms in Coconut

1. **Nitrogen deficiency** is mostly seen in young plants. Plant growth stops. Mature leaves turn yellow.
2. **Phosphorus deficiency** delays the emergence of spathes. Also, the nuts do not mature properly.
3. **Potash deficiency:** Especially in the lower leaves, pale green spots appear on the leaf blade on either side of the midrib, gradually enlarging and turning orange-yellow, then forming brown or rust-colored spots. The leaves gradually turn yellow. Trees are stunted, and leaves are small. Trees come to nut formation late. Nuts are few and small. Potash deficiency is the main reason for the formation of barren nuts and small nuts.
4. When important nutrients like **nitrogen and potash** are deficient in the soil, the flowers do not set properly, or the set nuts drop.

Details of Fertilizers to be Used in Coconut Orchards

Fertilizers should be applied in recommended doses from the year of planting. The use of organic manures like neem cake, farmyard manure, vermicompost, and green manure crops is very beneficial. Yields will be stable.

- **For trees aged 1-4 years:** 1/2 kg Urea + 1 kg Single Super Phosphate + 1 kg Muriate of Potash + 20 kg Farmyard manure/year/tree.
- **For nut forming trees older than 5 years:** 1 kg Urea + 2 kg Single Super Phosphate + 2.5 kg Muriate of Potash + 25 kg Farmyard manure or 2 kg Neem cake/year/tree.

Method of Applying Fertilizers

Only when fertilizers are applied correctly in the tree basin can they seep into the soil and be absorbed by the roots. Fertilizers should be applied in two equal splits, in June to July and September to October. A circular trench should be dug 3 feet away from the tree trunk, and apply fertiliser, covered with soil, and irrigated immediately. Adding salt to plants, cutting roots, etc., are not scientific methods. These actions harm the trees.

Inter-cultivation

The spaces between coconut rows should be cleanly plowed twice a year, once during the early monsoon and the second time after the rainy season. This helps control weeds. Additional income can be obtained by cultivating various intercrops in these spaces.

Damaging the roots of coconut trees by tilling too close to the base is harmful and does **not** increase yield. Roots are essential for water and nutrient uptake, and cutting them causes stress, weakens the tree, and makes it more prone to pests and diseases. This stress reduces flowering and fruit production, leading to lower yield and quality. Even if the tree survives, recovery takes time, affecting productivity for future seasons.

Role of Organic Manures in Coconut

The coconut plant absorbs many essential nutrients from the soil, for example, potassium, nitrogen, phosphorus, calcium, and magnesium in large quantities. Organic manures are very useful for stabilizing yields in coconut orchards. These manures decompose slowly in the soil and release nutrients. Farmers can easily prepare many types of organic manures in their own orchards with low investment. Research estimates that about 16 tonnes of organic matter (coconut leaves, spathes, husks, etc.) are wasted from one hectare of coconut orchard. This waste material can be converted into compost and made into excellent organic manure. While using organic manures, applying only 50 percent of the recommended chemical fertilizers is sufficient.

Benefits of Organic Manures

1. Helps retain moisture for a longer time, especially in light soils.
2. Increases the organic carbon content, which is an important component of the soil.
3. Increases the population of microorganisms in the soil, thereby increasing the availability of nutrients to the plant roots in the soil.
4. Reduces high levels of salts and alkalinity in the soil.
5. Improves the physical properties of the soil.

Using organic manures in coconut orchards every year gradually increases the available nutrients in the soil, helping the plants grow well. This also helps maintain soil health.

Organic Manures Useful in Coconut Orchards

1. **Vermicompost:** Vermicompost can be prepared from waste materials like dry leaves in coconut orchards using earthworms called 'Eudrilus eugeniae' (cow dung and leaves in a 1:1 ratio). 50 earthworms can be used per 10 kg of organic matter. In 2-3 months, the organic matter decomposes into excellent vermicompost. This compost manure contains about 1.8% nitrogen, 1% phosphorus, 0.16% potassium, etc.. In another method, compost can be made in the trench itself by digging a trench around the coconut tree, adding earthworms to the coconut leaves in the trench. In this method, the manure is ready in about 90 days.
2. **Coconut Coir Pith Compost:** Manure can be prepared from coconut coir pith using *Pleurotus* fungus powder (0.2%), lime (5%), urea (0.5%), rock phosphate (0.5%), and cow dung. The compost manure is ready in about 2 to 3 months. This manure contains about 1.2 to 1.8% nitrogen, 0.1 to 0.22% phosphorus, 0.1 to 0.4% potassium, etc.. This manure can be applied up to 20 to 25 kg per coconut tree.
3. **Green Manures:** Green manure crops like sun hemp, dhaincha, pillipesara, cowpea (alasande) should be grown as soon as the early monsoon rains fall and incorporated well into the soil after about 2 months. These green manure crops add major nutrients like nitrogen (6 to 7%), phosphorus (1 to 2%), and potassium (4 to 5%) to the soil. Besides these, many micronutrients also become available to the plants.
4. **Concentrated Organic Manures:** Concentrated organic manures like neem cake, pongamia cake, groundnut cake, etc., contain up to 8% nitrogen, 12% phosphorus, and potassium. These can also be used as organic manures at the rate of 25 kg per tree.
5. **Bulky Organic Manures:** Bulky organic manures like compost manure, farmyard manure, poultry manure, goat or sheep manure, etc., have lower nutrient values but offer many other benefits. These manures mix with the soil, make it friable, and allow more moisture to seep into the soil. This manure contains about 1 to 2% nitrogen, 0.5% phosphorus, and 1% potassium.

Straight Fertilizers vs. complex Fertilizers

1. **Straight Fertilizers:** These provide a single nutrient (e.g., Urea for Nitrogen, Superphosphate for Phosphorus, Muriate of Potash for Potassium). They are often more economical and allow for precise nutrient management based on the specific deficiencies identified in your soil or plant analysis. You can mix these straight fertilizers in the recommended proportions before application.

2. **"COMPLEX FERTILISERS" :** These contain two or more nutrients in a pre-mixed form (examples like 10-26-26, 12-32-16, and 15-15-15) While convenient, they might not always provide the exact ratio of nutrients your coconut need, potentially leading to over- or under-application of certain elements.

Farmers queries:

1. **Even Though i am applying the recommended dose of fertilisers, why did my coconut crop not improve immediately?**

In coconut crop the fertiliser applied in the soil should travel 30 to 40 feet height through xylem, phloem which needs at least 7 to 8 months to reach the top leaf located in the tree spindle. As you observe, coconut trees have 30 to 35 leaves and each freshleaf coming will bear a fresh panicle, so the problems in old panicles which are already opened and ready to set nuts will not be corrected immediately after application of fertilisers. As coconut is year round crop fertilisers should be applied in regular intervals matching the rains incase of rainfed instead of applying once. In irrigated gardens through applying fertilizers by drip one can ensure even distribution of fertilisers throughout the year.

2. **People told me to use biofertilizers in combination with chemical fertilizers in coconut gardens , how does it work?**

True, a combination of chemical and biofertilizers in coconut works wonders. The chemical fertilisers applied remain unavailable to plants if biofertilizers are not applied. The nitrogen , phosphorus,potash applied to plants in soil by chemical fertilizer application is to be converted to usable form and then only plants can take the nutrients applied which is only done by biofertilizers which carry this useful nitrogen,phosphorus, potash-fixing fungi. spending 10k on chemical fertilisers without using bio fertilizers gives only 4k to 5k results (40% to 50%)but spending additional 1k on biofertilizers in combination of 10k chemical fertilizers gives 20k benefit to farmers where the benefit doubles (200%).

3.Where will I get useful biofertilizers?

Biofactor, Hyderabad supplies the good strains of biofertilizers at reasonable cost.

4. **People are suggesting use of micronutrients like boron and magnesium along with chemical fertilisers why?**

Micronutrients play a critical role in coconut cultivation alongside straight fertilizers (like urea, superphosphate, and muriate of potash), as they support essential physiological functions that macronutrients alone cannot fulfill. While NPK fertilizers help in plant growth, root development, and nut setting, micronutrients such as Boron, Zinc, Magnesium, and Copper are vital for processes like enzyme activation, pollen viability, chlorophyll formation, and disease

resistance. Deficiencies can lead to issues like immature nut fall, button shedding, and poor quality copra.

For optimal growth and yield, the recommended micronutrient dosages per palm per year are as follows:

- Boron (as Borax): 50 to 100 g
- Zinc (as Zinc Sulphate): 100 to 200 g
- Magnesium (as Magnesium Sulphate): 500 to 1000 g
- Copper (as Copper Sulphate): 50 to 100 g
- Iron (as Ferrous Sulphate): 100 to 200 g
- Manganese (as Manganese Sulphate): 100 to 200 g

These should be applied in split doses (preferably twice a year) along with FYM or compost, and integrated with NPK schedules for balanced nutrition.

5. Excess button shedding in coconut is observed , why?

Button shedding in coconut palms is caused by: poor pollination, moisture stress, nutrient deficiencies (Boron, Potassium, Calcium), pests/diseases (red palm weevil, rhinoceros beetle, fungal infections), high temperature/drought, waterlogging, imbalanced fertilizer use (excess nitrogen), and genetic factors.

Solutions include: proper irrigation, balanced fertilizers (NPK 500:320:1200g/tree/year; 200g Borax/tree/year), foliar sprays (0.5% Borax, 1% Potassium Sulphate), pest/disease management, mulching, growth regulators (NAA 20 ppm), promoting pollination, and drainage management.

6. Chicken Manure vs. Sheep Manure for Coconut Trees ,which is best for coconut gardens?

Both chicken manure and sheep manure are excellent organic fertilizers, but they have slightly different nutrient profiles:

- **Chicken Manure:** Generally richer in nitrogen (N), phosphorus (P), and potassium (K) compared to other common farm manures. It's often considered a "hot" manure due to its higher nutrient content and faster decomposition. It can quickly boost plant growth.
- **Sheep Manure:** Also a good source of N, P, and K, along with various micronutrients. It tends to be less "hot" than chicken manure, decomposing more slowly and providing a more gradual release of nutrients. It also improves soil structure effectively.

Both are highly beneficial. The "best" choice often depends on availability and cost.

- If you need a quicker nutrient boost, chicken manure might be slightly preferred due to its higher immediate nutrient availability.
- For long-term soil health improvement and a steady supply of nutrients, sheep manure is excellent.

Benefits of Intercropping in Coconut

1. There are many benefits to intercropping in coconut. Generally, like other commercial crops, prices in the coconut market fluctuate. Due to the lack of significant development in coconut-based industries and changes in coconut oil prices domestically and internationally, coconut prices fluctuate. Therefore, instead of cultivating coconut as a monocrop, farmers can earn an additional income of about Rs. 20,000 to 25,000/- per acre along with coconut through intercropping.
2. Through intercropping, coconut yields and income remain stable.
3. About 4 to 5 meters of space between coconut rows remains vacant, allowing for the growth of many weeds. Therefore, intercropping can help control these weeds.
4. Natural resources like land, water, and sunlight can be utilized efficiently.

Details of Annual and Perennial Crops

- **Annual Crops:** In young coconut orchards less than 5 years old, there is good air and light penetration. Under these conditions, until the coconut trees come into full bearing, annual commercial crops like turmeric and ginger, vegetables like yam, colocasia, cabbage, cauliflower, beans, vine vegetables, chili, brinjal, okra, pulses, and groundnut can be grown.
- **Perennial Crops:** In coconut orchards older than 20 years, perennial crops like banana, cocoa, pepper, pineapple, papaya, or commercial flower plants like heliconia can be grown. Through these, a stable income can be obtained along with coconut within 2 to 3 years of planting.

In high-density coconut cropping system experiments, applying 75% of recommended fertilizers (Nitrogen, Phosphorus, Potash) along with vermicompost made from the waste materials of the respective intercrops (banana, cocoa, heliconia, yam/tomato, pineapple) resulted in higher yields. Analyzing the cost-benefit ratio, it was determined that a profit of Rs. 3.22 can be obtained for every rupee spent.

In light soils like red gravel and sandy soils with limited water facilities, moisture is retained in the soil only during the rainy season. Therefore, fast-growing crops like beans, groundnut, and

maize can be cultivated during the early monsoon. In rain-fed orchards, growing green manure crops like sunn hemp, pillipesara, dhaincha, etc., can enrich the soil and improve water infiltration. While applying fertilizers to coconut, additional fertilizers based on the intercrop must also be provided. If farmers do not take this precaution, there is a possibility of reduced coconut yield.

In coconut orchards of coastal Andhra Pradesh, intercropping with "medicinal" and aromatic crops like patchouli (2.84 profit-cost ratio) and citronella (2.12 profit-cost ratio) has been identified as profitable.

suggest best intercrops in coconut as per the age group of coconut plantation

Intercropping in a coconut plantation is a fantastic way to maximize land use, diversify income, and improve soil health. The best intercrops depend largely on the age of your coconut trees, as light availability changes significantly over their lifespan.

Plant Protection

Pests

Among the pests that attack coconut, the black-headed caterpillar, rhinoceros beetle, red palm weevil, leaf caterpillar (slug caterpillar), spiraling whitefly, and eriophyid mite are important.

Black-headed Caterpillar

The black-headed caterpillar infests coconut trees and orchards of all ages (nurseries, mature orchards, bunds of paddy fields, banks of fish ponds). It enters the lower fronds and scrapes and eats the chlorophyll. It builds nests with frass resembling sawdust and silk threads, sticking the leaflets together, and resides within these nests.

Symptoms/Damage caused by Black-headed Caterpillar:

- Dried patches appear on the infested leaves.
- As the infestation increases, the patches merge, and the entire leaf dries up.
- Thus, all the leaves on the tree dry up and droop.
- The entire orchard looks burnt. In severe infestations, the nuts can also be damaged.
- When young plants are infested, they become stunted and come to nut formation late.
- When older plants are infested, the chlorophyll responsible for food production in the leaves is eaten away, reducing the food supply to the plant, leading to reduced yield and nut size.

- Due to the drying of leaves, young nuts drop. Spathes fail to emerge, reducing yields. It takes 2 to 3 years for the tree to recover.

Life Cycle: The adult moth, with greyish wings, lays eggs near old galleries on the lower fronds. The eggs hatch in 5 days, and the small caterpillars start feeding on the leaves. The caterpillars feed within the leaf folds for about a month. They pupate within cocoons made of frass, and the pupal stage lasts 10 to 12 days, after which the adult moth emerges. The life cycle is completed in about six weeks.

Management:

- Infested and dried lower leaves should be removed and burnt.
- If the infestation is low (pest present on 1 or 2 leaves), release the parasitoid *Goniozus nephantidis* (10 individuals) and *Bracon brevicornis* (hebetor) (20 individuals) per tree on 10 percent of the trees in the orchard. If the black-headed caterpillar infestation is high, a larger number of parasitoids need to be released.
- Since this pest also infests palmyra trees in the vicinity of coconut orchards, parasitoids should also be released on these trees.
- Infested and dried orchards recover faster and yield sooner if recommended fertilizers (1 kg Urea, 2 kg Super Phosphate, and 2.5 kg Muriate of Potash/tree/year) are applied and irrigated.

Rhinoceros Beetle

The adult rhinoceros beetles are robust, black beetles with hard wings. They are called rhinoceros beetles because they have a horn on their head like a rhinoceros.

Symptoms (Rhinoceros Beetle): Unopened, mature leaves show 'V' shaped cuts. Boreholes made by the beetle in the spindle, and chewed fibrous material in the spindle are visible.

Damage:

- The rhinoceros beetle bores into the tender spindle portion and spathes, damaging unopened young leaves and spathes, thus reducing coconut yields.
- In young plants, infestation can damage the spindle, potentially killing the plant.
- Besides the direct damage, the beetle spreads bud rot disease as it moves from tree to tree.

- Furthermore, the fermented smell emanating from the damaged spindle attracts the Red Palm Weevil, which can cause further damage.

Life Cycle: White, oval eggs are laid individually 2 to 6 inches deep in moderately moist areas and covered with surrounding material. By the time they hatch, the eggs swell and change color to yellow-brown. Eggs hatch into larvae in 7 to 20 days. A single female beetle lays about 140 eggs in 6 to 7 batches. Eggs are laid in decaying organic matter in the orchard, dead coconut trees, stumps, etc.. The grub that hatches from the egg is white with hard parts. It feeds voraciously on organic matter and reaches the pupal stage in 99 to 180 days. To pupate, it creates a smooth chamber and turns into a reddish-brown pupa. The adult beetle emerges from the pupal stage in 17 to 62 days. It remains in the chamber for about 10 days after emerging before coming out. It then moves around the spindles of coconut trees or breeding sites like dead coconut trees, manure heaps, or compost pits, living for more than 200 days. The life cycle is completed in about 130 to 196 days. These beetles are active throughout the year, causing damage to trees.

Control:

- Remove fallen trees from the orchard and utilize or destroy them appropriately.
- Dig out and burn the stumps of fallen trees (the part remaining in the ground).
- Do not keep cattle manure heaps in the orchard. Turn nearby cattle manure heaps every 3 months and spray with a solution of *Metarhizium anisopliae* @ **500 billion** spores/cubic meter (5 g/L of water) to thoroughly drench the heaps.
- Grubs also thrive well in vermicompost pits. Therefore, cover these pits with nylon nets to prevent adult beetles from laying eggs. Release rhinoceros beetles infected with Baculovirus disease into the orchard at the rate of 10 or 15 per hectare every three months.
- Every 3 months, apply a mixture of 100 grams of neem cake or neem seed powder mixed with 150 grams of sand in the spindle and leaf axils of coconut trees.
- In young plants where rhinoceros beetle infestation is severe, place 5 g or 10 g of “Chlorantraniliprole 4G” granules in small polythene packets. Make small holes in these packets and place at least 4 packets per tree in the crown area.
- Set up pheromone traps (Rhinolure) in buckets at the rate of one trap per five acres to attract and kill adult beetles.
- To reduce rhinoceros beetle severity, place naphthalene balls (12 g/tree, i.e., 2 balls) in the leaf axils near the crown of the coconut tree.
- Trap Type: Commonly used traps are bucket traps.

Framer frequent quires:

- 1. Eventhough i am using more no of pheromone traps in my coconut field the infestation of the rhinoceros beetle and red palm weevil is not controlled why?**

It is observed that the farmers are using more number of pheromone traps per hectare area without following proper guidance from company or scientists, which is increasing the pest population attracting from side by fields also. The recommendation is 1 trap per hectare for continuous land, if the field is located in lands less than 1 hectare, put 1 trap per one land of the field.

- 2. I am using pheromone traps for controlling rhinoceros beetles and red palm weevils but why aren't pests controlled?**

It is observed that 1. placing the traps wrongly, tying to tree 2. The lure veil having holes should be pinned to get the chemical smell released properly 3. placing the trap at above or below the pest fleet height 4. not changing the lure after the expiry date with new one 5. not changing the water in the bucket as the dead insects smell stops attracting further catches 6.using correct lure for the designated pest using red palm lure will not catch rhinoceros or either.

- 3. Where will I get the pheromone traps for coconut?**

You can buy pheromone traps for coconut from Agri phero solutions, Hyderabad. They have developed this in collaboration with CPCRI using nanotechnology which slowly releases the chemical on a daily basis which increases the daily catches.

- 4. What is the price of the coconut pheromone traps?**

The cost of the bucket which is used for both Rhinoceros beetle and red palm weevil is Rs.160/-

The cost of the Rhinoceros lure is Rs.250/-

The cost of the Red palm weevil lure is Rs.120/-

- 5. For control of Rhinoceros beetle pouring mixture of sand and neem cake in the spindle is recommended why?**

Every 3 months, apply a mixture of 100 grams of neem cake or neem seed powder mixed with 150 grams of sand in the spindle and leaf axils of coconut trees.

Mixture of sand and neem cake is recommended for control if redpalm weevil because 1.the smell of neem cake repels the pest for some days 2. When the insect comes to effect the new plantation when the insects chews the young leaves of the spindle it will hinder its mouth movement not only that in the already affected plants the pest feels suffocated .4.The azadirachtin chemical releasing slowly in to the plant will act as antifungal treatment and control the spread of further fungal diseases as a result of post rhinoceros beetle infestation.

6. For control of Rhinoceros beetle naphthalene balls placement in the leaf axils is suggested,why?

Putting naphthalene balls in the leaf axils is recommended to repel the pest by the strong smell released by naphthalene balls continuously. place naphthalene balls (12 g/tree, i.e., 2 balls) in the leaf axils near the crown of the coconut tree.

7. For controlling Rhinoceros beetle spraying chemical or application granules will work or not?

For controlling Rhinoceros beetles, spraying chemicals will not work effectively as the active adult pest dwells deep in the coconut spindle and younger ones live in the compost heaps and dead coconut trees in the field. Placing granules directly releases the smell immediately at once which stops the pest for just 2 or 3 days, but placing 5 g or 10 g of Chlorantraniliprole 4G granules in small polythene packets. Make small holes in these packets and place at least 4 packets per tree in the crown area.small holes to the pouches will release the smell slowly and repel the pest for more no of days effectively. 4 pouches per tree to be placed in the leaf axils .

8. Suggested removal of dead coconut and palm trees in my coconut field to control Rhinoceros and red palm weevil , why?

The dead coconut of palm trees act as sources of breeding for the Rhinoceros beetle and Red palm weevil hence the removal of dead and diseased coconut and palm trees periodically is recommended

9. Suggested not to throw cow dung or tie cattle in the coconut fields, why?

It is observed that the undecomposed cow dung and its heaps in and around coconut fields acts as breeding sites to the Rhinoceros beetle and affects the coconut plantation. It is also suggested not to apply undecomposed farm yard manure to the coconut plants . Do not keep cattle manure heaps in the orchard. Turn nearby cattle manure heaps every 3 months and spray with a solution of

Metarhizium anisopliae @ 5×10^{11} spores/cubic meter (5 g/L of water) to thoroughly drench the heaps.

- Grubs also thrive well in vermicompost pits. Therefore, cover these pits with nylon nets to prevent adult beetles from laying eggs.
- Release rhinoceros beetles infected with Baculovirus disease into the orchard at the rate of 10 or 15 per hectare every three months.

Red Palm Weevil

Damage from the Red Palm Weevil is very severe. If neglected, the entire orchard can be destroyed. Orchards aged 5 to 10 years are often most susceptible to this pest. This pest is also more common in orchards where banana is grown as an intercrop. Furthermore, the Red Palm Weevil is attracted to trees that have died due to bud rot or lightning strikes, or trees whose spindle have been infested by the rhinoceros beetle. The weevil is also attracted to wounds on the stem caused by tools, root pruning, or damage to the base during plowing with a tractor.

Symptoms:

- Leaves in the middle whorls turn yellow or dry up.
- Thick, dark red gum oozes from boreholes on the stem. Chewed fibrous material comes out of the holes.
- The base of the frond splits vertically.
- Cocoons made of fiber by the grub, or chewed fibrous material, are visible in the leaf axils or at the base of the tree.
- If you place your ear against the stem of an infested tree, you can hear the sound of the grubs inside eating the stem.

Damage:

- The pest that causes a vertical splitting at the base of the stem in coconut cultivation is the **Red Palm Weevil** (*Rhynchophorus ferrugineus*).
- When the tree is partially damaged, growth is reduced, and nut formation is delayed.
- When the trunk becomes completely hollow, the tree dies and falls over.

Life Cycle: Adult weevils lay about 200 smooth, shiny, brownish eggs individually in cracks, cuts, or wounds on the stem. Within 2 to 5 days, legless grubs hatch from the eggs and feed on

the soft tissue inside the stem. The grubs are spindle-shaped with wrinkled skin, bulging in the middle, thick, brownish, legless, and have a dark red head. They feed for two to three weeks, then pupate, and emerge as adult weevils. The life cycle takes about 3 months to complete. The adult weevil lives for 2 to 3 months.

Management:

- Immediately cut down, split open, and burn trees killed by this pest.
- Immediately remove trees killed by lightning or bud rot from the orchard.
- In spindle infested by rhinoceros beetles, remove the beetle and fill the hole with a mixture of 100g neem cake + 150g sand.

Do not make any kind of wounds on the stem or base of the coconut tree. Do not cut the roots or pull green leaves. Place the artificial lure 'Pherro lure' in a bucket to attract Red Palm Weevil adults in groups. If arranged at a height of 1.5 meters on the coconut tree trunk, the Red Palm Weevil adults will be attracted, fall into the poison bait inside the bucket, and die. Mix 1 ml of Imidacloprid insecticide per liter of water and pour it into the part bored by the Red Palm Weevil on the tree trunk. Clean the debris and rotten material from the half-hollowed trunk and apply coal tar or Japan black as a coating on the wound. If the tree cavity is filled with a mixture of cement, sand, and stones, the tree will be strong and sturdy, preventing it from falling during strong winds.

- **Trap Type: Bucket traps** are commonly used for Red Palm Weevils.

By implementing these pheromone trapping strategies, coconut farmers can significantly reduce the impact of these destructive pests and improve their crop yields.

Farmer queries:

1.Young coconut trees of age 5 and above years while formation of bole falling and dying?

Coconut orchards aged 5 to 10 years are often most susceptible to red palm weevil pest. This pest is also more common in orchards where banana is grown as an intercrop. Furthermore, the Red Palm Weevil is attracted to trees that have died due to bud rot or lightning strikes, or trees whose spindle have been infested by the rhinoceros beetle. The weevil is also attracted to wounds on the stem caused by tools, root pruning, or damage to the base during plowing with a tractor.

Eriophyid Mite

Among the pests that infest and cause damage to coconut in our state, the eriophyid mite is significant. Although other mites besides coconut-damaging insects have been previously recorded, the eriophyid mite *Aceria guerreronis* has been recorded on coconut nuts in our country.

Symptoms:

- This mite predominantly infests tender nuts that are usually 2 to 3 months old.
- These mites gather in thousands under the perianth (calyx/bracts or 'muchika') near the soft growing tissue and suck the sap, initially causing elongated white streaks.
- Later, yellow or brown triangular patches spread from the perianth towards the lower side of the nut.
- Gradually, these patches turn reddish-brown, the skin dries up, develops vertical cracks, and the nuts become scarred (like 'gajji kayalu' - scabbed nuts).
- Sometimes gum oozes from the cracks.

Damage:

- Infested young nuts drop off. Coconut nut growth is stunted, leading to reduced size.
- Due to the reduction in nut size, the coconut yield decreases by about 25 percent. Both the yield and quality of copra decrease.
- The coir on mite-infested nuts hardens, making it unsuitable for the coir industry. Mite-infested nuts do not fetch a good price in the market.

Life Cycle: This mite is brownish and resembles a thin thread. It has two pairs of legs towards the head. This mite is visible only under a microscope. These mites, measuring 200-250 microns in length and 40 microns in thickness, live in colonies of thousands under the perianth of the young nut. They also live near the ovary in the coconut flower. The female mite lays 20 to 100 eggs. The eggs hatch into nymphs in 7 to 10 days. If the mite population under the perianth becomes very high, they settle on the surface of the nut or young nut. From there, these mites spread to other trees through the wind. These mites also spread from one place to another through rain and other insects. The mite also spreads when infested nuts are transported from one area to another.

Management:

- Destroy fallen coconut young nuts and nuts infested by the mite.
- Apply neem cake at 5 to 10 kg/tree/year. Along with neem cake, use other organic manures and green manures extensively.

- When mite infestation is severe, apply additional Muriate of Potash up to one kg. This means applying up to three and a half kilos of Muriate of Potash per tree.
- Provide irrigation regularly.
- In fertile lands, reduce mite infestation by growing intercrops like banana, yam, cocoa, turmeric, or vegetables.
- Mix Azadirachtin 10000 ppm (a neem-based product) at 5 ml/liter of water and spray on the bunches, OR administer 10 ml of the product + 10 ml of water per tree through root feeding. This should be done three times a year.

Farmers queries:

1. People are recommending more potash applications to control mite infestation in coconut , why?

When mite infestation is severe, apply additional Muriate of Potash up to one kg. This means applying up to three and a half kilos of Muriate of Potash per tree. This will make the nut's collar region too hard early as mites attach this area of the nut during development applying potash will harden the nut and reduce mite infestation.

2. people suggested application of neem cake and other organic manures, green manures regularly to coconut orchards , why?

Applying neem cake at 5 to 10 kg/tree/year will control soil borne pests and diseases in coconut where soil around the tree basin acts as place of survival for nymphs of mites, ganoderma spores and other useful fungi. Use of neem cake regularly encourages useful fungal growth and at the same time arrests the growth of harmful fungi like Ganoderma and nymphal stages of mites .Along with neem cake, use of other organic manures and green manures extensively encourages beneficial soil microorganisms.

Slug Caterpillar (Leaf Caterpillar/Aku Telu)

The infestation of the slug caterpillar (aku telu) is high during summer. This pest develops rapidly under dry weather conditions. Heavy dew in the morning followed by high daytime temperatures increases the slug caterpillar infestation.

Symptoms:

- Young caterpillars scrape and eat the chlorophyll on the underside of the leaf, causing round patches on the leaves.
- As the caterpillars grow, they eat the leaf blade from the edges, leaving only the midribs. Dried leaves hang from the tree.

- In severe infestations, they also attack branches and nuts.
- Frass (excreta) excreted by the caterpillars forms a layer under and around the tree.
- When the infestation is severe, the slug caterpillar also attacks and destroys intercrops like banana and cocoa.

Management:

- Monitor the presence of adult moths using light traps at the end of January and take appropriate control measures.
- To control adult slug caterpillar moths, place 3 bulbs of 200 watts per hectare (two and a half acres) at a height of one and a half feet above the ground, with a water trough below, to attract the moths.
- Hundreds of slug caterpillars pupate in the fibrous material in the crown and leaf axils of the coconut tree. Therefore, clean the spindle to destroy the pupal stages.
- Cut and burn infested and dried leaves. This destroys the pupae in the dormant stage.
- When adult moths are present in the orchard, use light traps to attract and eliminate large numbers of slug caterpillar moths.
- When slug caterpillars are in the larval stage, release the parasitoid *Pediobius imbrues* at the rate of 60 per tree every 15 days.
- Spray Chlorantraniliprole 18.5% SC at 0.3 ml/liter. If the infestation persists, spray B.t. (*Bacillus thuringiensis*) at 5 g/liter.
- Trees recover quickly if irrigated properly and fertilized with the recommended dose (1 kg Urea, 2 kg Single Superphosphate, 2 kg Muriate of Potash per tree).

Spiraling Whitefly

The scientific name of the Rugose spiraling whitefly is *Aleurodicus rugioperculatus*. It was first reported infesting coconut in 2016 in Pollachi, Tamil Nadu, and Palakkad, Kerala. In Andhra Pradesh, this spiraling whitefly was first identified on coconut trees in the Kadiyam region of East Godavari district in December 2016.

Symptoms:

- It is a type of sap-sucking winged insect.
- It lays eggs in white spiral patterns on the underside of leaves. These spirals are covered with a white waxy coating and hairs.

- The underside of the leaves becomes covered with a white cottony substance.
- Honey-like sticky substance drips from the underside of the leaves onto the leaves below, causing them to be covered with black sooty mold.

Damage:

- Both the nymphal and adult stages of the whitefly gather on the underside of coconut leaves and suck the sap. This causes stress to the trees.
- The honey-like sticky substance promotes the growth of a black sooty mold fungus belonging to the genus *Capnodium* (specifically mentioned as *Leptoxiphium*).
- The sooty mold on the leaves obstructs photosynthesis.
- In severe whitefly infestations, not only the leaves but also the petiole bases and young coconuts get covered with the white cottony substance.

Life Cycle: The adult female lays light brown or dark yellow eggs in a spiral pattern on the underside of the leaves. This whitefly develops through five stages. The spiraling whitefly is about five times larger than the common whitefly (approx. 2.5 mm). The wings of the spiraling whitefly have irregular light brown bands spread across them. Its life cycle is completed in about 40 to 45 days. The Rugose whitefly becomes severe under drought conditions. They cannot tolerate heavy rain and humid weather. An increase in temperature by two degrees Celsius or more above the normal summer temperature contributes to the whitefly outbreak.

Integrated Management of Spiraling Whitefly:

- Do not transport coconut seedlings from severely infested areas to other regions. If export is necessary, inspect thoroughly to ensure the absence of whitefly before transportation.
- Tie 12 yellow tarpaulin sheets per acre to the coconut tree trunks and apply crude castor oil to these sheets. The adult spiraling whiteflies get stuck to the sticky tarpaulin sheets. Reapply castor oil to the yellow sheets every 7 to 10 days.
- The natural enemy, the parasitoid *Encarsia guadeloupae*, keeps this whitefly under control. Since there is no facility for laboratory production of this parasitoid, collect it from coconut orchards where *Encarsia* has naturally developed and release it in whitefly-infested areas lacking the parasitoid. These parasitoids bring the whitefly under control in about 6 to 8 months.
- Since the Rugose whitefly is an invasive alien insect in our country, identify it and provide complete information to farmers and horticulture department officials. Provide appropriate training for its control.

- Stop the use of insecticides in orchards infested with spiraling whitefly. This allows natural enemies like parasitoids to multiply and control the whitefly.
- If the pest level is high, spray Azadirachtin 10000 ppm at 1 ml/liter OR 0.5% neem oil mixed with 10 g Surf per liter of water, ensuring thorough coverage of the undersides of the leaves.
- When the spiraling whitefly infestation is low, release 100-150 eggs of the predator *Chrysoperla zastrowi sillemi* (referred to as *Apertochrysa (Chrysoperla) zastrowi sillemi* / *aster* in the text) per tree; for moderate infestation, release 300 to 500 eggs per tree. These predators are available at Dr. Y.S.R. H.U. - Horticultural Research Station - Ambajipeta.
- Mix the entomopathogenic fungus *Isaria fumosorosea* (NBAIR strain Pfu5 mentioned) at 5 g/liter of water (1×10^8 spores/ml) and spray on the leaves.
- A bio-intensive approach effectively controls this harmful whitefly.

Sooty Mold Management:

- The black sooty mold is a secondary issue caused by the honeydew.
- **Water spray:** Applying pressurized water can help wash away the sooty mold.
- **Starch/Maida paste solution:** Spraying 1% starch solution or 2.5% maida paste solution mixed with detergent/Khadi soap (5g/liter) can help dislodge the sooty mold growth, allowing leaves to photosynthesize effectively.

Disease Management

Among the diseases affecting coconut, Ganoderma (Basal Stem Rot/Red Lac), Stem Bleeding, Bud Rot, Nut Fall, and Leaf Blight caused by fungal species are important.

Ganoderma Disease (Basal Stem Rot)

Ganoderma disease is a major and very dangerous disease affecting coconut orchards. It is also known as 'siga tegulu', 'erralakka tegulu' (red lac disease), 'bankakaru tegulu' (gum oozing disease), 'pottulakka tegulu' (husk lac disease). This disease is more prevalent in coconut orchards on light soils than on black soils. Its spread is also extensive in light soils. It is observed that the disease spread is less in years with high rainfall. Generally, weather conditions from November to June favor the spread of this disease. The severity and spread of this disease are higher in orchards experiencing significant water stress. Farmers commonly practice cutting coconut tree roots, which should not be done. Cutting coconut roots creates wounds, allowing the Ganoderma fungus spores present in the soil to infect the trees through the wounded roots.

Symptoms: Ganoderma disease infects coconut through fungal spores present in the soil. Initially, Ganoderma attacks the roots in the ground. At this stage, we cannot detect the disease. After a significant percentage of roots rot, the disease spreads into the stem, causing complete decay of the tissues. However, this rot is internal within the ground and stem, so it is not visible externally. Gradually, the disease progresses from the roots into the stem. At this stage, oozing of a thick, gummy liquid, dark brown to whitish in color, can be observed from small cracks around the base of the stem. Due to the complete decay of the stem tissue, the tree cannot absorb necessary water and nutrients from the soil. At this stage, gum oozes first through small cracks around the base of the stem. This gum oozing gradually spreads upwards. The leaves of the infected tree turn yellow, wilt, and become brownish. Due to the delayed emergence of new leaves, the number of leaves on the diseased tree is noticeably reduced. Similarly, the leaf size also decreases. In Ganoderma-infected coconut plants, the number of flowers significantly decreases, the number of male flowers increases, and the number of female flowers drastically reduces. Shedding of young nuts and nuts is also a symptom of this disease. If the roots of an infected coconut tree are observed, they appear completely rotten and turned black. In severely affected trees, about 70 percent of the roots are rotten. New roots do not develop. If neglected even slightly, the disease intensifies, the tree leaves turn yellow, wilt, and reach a drooping stage. Later, the entire crown wilts, and the tree dries up. This symptom should be identified as Ganoderma disease. Subsequently, these dried coconut leaves fall off. Due to the complete shedding of coconut leaves, the stem becomes bare and looks like a sharpened pencil tip. The tree dies within the next 5 to 6 months. Sometimes, mushrooms sprout on the bases of trees in the final stages of the disease. The upper surface of these mushrooms is lac-colored. The lower surface is white or greyish. If this part is observed carefully, millions of microscopic pores are visible. From these, billions of disease-causing fungal spores are released and spread through the air. When wet, these germinate, grow the fungus, and infect nearby coconut roots. Therefore, allowing Ganoderma disease to progress until mushrooms form is extremely dangerous. We often observe Ganoderma-infected dead trees breaking and falling at the base during strong winds in coconut orchards. When examining these fallen coconut logs, the inner part of the log is found to be rotten and hollow.

Integrated Management Practices including Biological Control for Ganoderma Disease:

- As soon as Ganoderma symptoms are observed in the orchard, isolate the affected area by digging a trench 1 meter deep and 50 cm wide around the diseased tree(s). This prevents the fungal spores near the roots of the infected trees from reaching healthy trees.
- Uproot dead and severely infected trees along with their roots and burn them. Otherwise, mushrooms will sprout on the dead tree stems, making disease control difficult. Burn

trash and grass in those pits. Under no circumstances should the stumps of dead trees be left in the orchard, as newly planted seedlings may die.

- Before replanting seedlings in the place of dead trees, expose the pits to sunlight for a week to ten days to dry thoroughly. Then, burning trash in the pit can help control root rot fungi.
- When replanting new seedlings, fill the pits (where trash was burnt) with well-decomposed farmyard manure, compost, and a mixture of 50 g of *Trichoderma viride* fungal powder mixed with 1 kg of neem cake before planting the seedling.
- Ganoderma fungal spores reside in the soil and infect trees through cut or damaged roots. Therefore, avoid cutting coconut tree roots and deep plowing that might sever roots. Under no circumstances should tree roots be cut.
- This disease occurs more frequently in light soils. Therefore, in these light soils, grow green manure crops like dhaincha and sunn hemp densely and incorporate them into the soil at the 50% flowering stage. This increases organic matter in the soil and enhances its water retention capacity. Additionally, fungi that control the Ganoderma fungus also multiply.
- Make basins with a radius of two meters around the base of the trees and irrigate each tree individually through channels/furrows. Irrigating the orchard in beds allows the Ganoderma fungus to spread to all trees through water.
- In soils with Ganoderma disease, apply 50 g of *Trichoderma viride* (or *Trichoderma asperellum* mentioned as alternative/synonym) fungal powder mixed with 5 kg of neem cake to each coconut tree annually. This allows *Trichoderma* to multiply well in the neem cake and suppress the development of Ganoderma-causing fungal spores.
- In soils with high Ganoderma infestation, apply 125 g *Trichoderma viride* (*Trichoderma asperellum*) + 125 g *Pseudomonas fluorescens* fungal powder mixed with 5 kg of neem cake once a year per tree. (Note: Dosage differs slightly here from previous point, suggesting higher application for severe cases).
- If Ganoderma symptoms appear on even a single tree in the coconut orchard, take the measures described above for all trees in that orchard.
- Do not mix *Trichoderma* fungal powder with Bordeaux mixture or other fungicides. Doing so will kill the *Trichoderma* fungus due to the effect of those chemicals. Therefore, under no circumstances should Bordeaux mixture be used before or after applying *Trichoderma viride* (*Trichoderma asperellum*). Besides this detrimental effect, using Bordeaux mixture also destroys naturally occurring *Trichoderma* and other beneficial microorganisms in the soil.

Farmers queries:

1. People are suggesting application of *Trichoderma viridae* paste on the stem and soil application along with neem cake or any oil cake for control of *Ganoderma* in coconut, why?

Trichoderma viridae is a beneficial fungi which controls the spread of *Ganoderma* in the soil and also on the affected portions of the coconut. Soil application should be done with neem cake or any oil cake or any biological matter like Farm yard manure which ensures the establishment of *Trichoderma viridae* in the soil in the initial stages, when applied directly without any organic matter *Trichoderma* can not thrive in the harsh climate and die. It should not be mixed with any other fungicides or insecticides also.

2. Even Though after application of *Trichoderma viridae*, *Ganoderma* is not controlled in my coconut garden, why?

The possible reasons for *Trichoderma viridae* not worked for *Ganoderma* control in coconut may be

a) Direct soil application without using a organic material, Soil application should be done with neem cake or any oil cake or any biological matter like Farm yard manure which ensures the establishment of *Trichoderma viridae* in the soil in the initial stages, when applied directly without any organic matter *Trichoderma* can not thrive the harsh climate and die.

b) Mixing *Trichoderma* with any fungicide like bordaux or chemical will result in death of the fungus. It should not be mixed with any other fungicides or insecticides also.

c) Quality or spore count of the *Trichoderma* procured, when the spore count is not up to the mark it cannot multiply to that level to fight and control *Ganoderma* disease. sourcing highly quality material from trusted source is the best solution

d) Lack of organic carbon in the soil, coconut gardens established in sandy loams often face this problem, as organic carbon is not maintained in the soil the useful fungi like *Trichoderma viride*, *Pseudomonas* etc.. can not thrive and sustain for the long term. To avoid this, apply organic manures, oil cakes frequently instead of chemical fertilisers, practice intercropping and make green manuring a regular practice to improve soil health.

3. Where will I get good quality *Trichoderma viride*?

You can avail biofactor “Invictus” near by dealer shop or online, price Rs130/-kg.

Stem Bleeding Disease (Nalla Macha Telugu)

Stem bleeding disease is also called black scorch disease (nalla lakka tegulu). This disease is caused by the fungus *Thielaviopsis paradoxa*. This disease affects coconut trees of all ages but spreads more rapidly on younger trees. High water table and extreme soil pH (acidic or alkaline) contribute to this disease.

Stem Bleeding Symptoms: The main symptom is the oozing of a dark liquid from cracks on the stem. The oozed liquid turns black, forming a black patch. The entire portion beneath this black patch rots. If this black patch area is chipped, a yellow to brown liquid comes out. As the disease progresses, the bark on the stem peels off. When conditions are favorable for the disease, the entire stem appears covered in black streaks. Coconut yield decreases in trees affected by this disease. When this disease attacks young plants, there is a risk of the plants dying as the fungus enters the tree through wounds or cracks on the stem and causes the disease.

Stem Bleeding Management Practices:

- Avoid causing any injury to the coconut tree stem.
- As soon as disease symptoms appear on the stem, prepare a paste of *Trichoderma viride* (*Trichoderma asperellum*) fungal powder and apply it to that area (a paste is formed by mixing 50 g powder with 25 ml water).
- The fungal spores causing this disease, similar to *Ganoderma* spores, reside in the soil and infect coconut through natural cracks on the stem under suitable environmental conditions. Therefore, to control the fungal spores in the soil, it is beneficial to apply 50 g of *Trichoderma viride* powder mixed with 5 kg of neem cake in the tree basins every year.
- As soon as stem bleeding symptoms appear on the stem, knife out the affected part and apply *Trichoderma viride* (*Trichoderma asperellum*) paste over it. Doing this allows *Trichoderma* spores to develop and thereby control the stem bleeding disease.

Bud Rot Disease (Movvu Kullu Tegulu)

Bud rot and nut rot diseases attack and damage coconut during the rainy season. Disease symptoms are likely to appear with the onset of rains.

Bud Rot Symptoms: In infected plants, the spear leaf (central unopened leaf) and the two or three adjacent leaves initially wilt. Later, other secondary organisms invade, causing the bud to rot completely, emitting a foul odor. If this leaf is pulled, it comes off easily. This rot spreads below the spear leaf and attacks the single growing point (apical bud) of the coconut tree, eventually killing the bud and the tree. Sometimes, even though the bud rots, the growing point survives, and new leaves emerge after a few months. However, these leaves are often stunted, malformed, with burnt edges. Such leaves

remain compressed by the fibrous material in the crown, appearing bushy. If control measures are not taken in time, this disease can cause severe damage to coconut orchards.

Precautions for Bud Rot Control:

- Farmers should take preventive measures before the disease occurs rather than taking control measures after infection.
- Ensure proper drainage in the orchard to prevent water stagnation.
- Remove and burn trees killed by the disease.
- On trees affected by bud rot, remove the rotten bud and the adjacent rotten parts and burn them. On trees with stunted, short leaves, cut away the fibrous material in the crown to allow leaves to emerge easily.
- Applying potash fertilizers at the recommended dose regularly increases disease resistance in plants.
- Remove bunches affected by nut rot and spray a copper fungicide (Copper Oxychloride) solution (3 g per liter of water) ensuring thorough drenching of other bunches and the crown area.
- In orchards subjected to flooding, wash away the silt deposited in the spindle thoroughly as soon as the flood water recedes. Then, spray 0.3% Copper Oxychloride solution (3 g per liter of water), ensuring complete coverage of the crown and leaves.

Biological Control Method:

- Apply *Pseudomonas fluorescens* talc powder to the crown region of the coconut plant. Based on plant age: 5 grams for plants less than a year old, 10 grams for one-year-old plants. Similarly, for plants aged 2, 3, 4, 5 years and older, apply 75, 100, 150, 200 grams of talc powder respectively.
- Remove bunches affected by nut rot and spray *Pseudomonas fluorescens* culture solution, thoroughly drenching other bunches and the crown area.
- If bud rot occurs in nursery coconut seedlings, applying *Trichoderma harzianum* (referred to as *Trichoderma reesei* in text, likely a typo or specific strain used) talc powder (5 g per seedling) to the crown region can control bud rot in the nursery.
- In flood-affected orchards, after the water recedes, thoroughly wash away the silt from the spindle. Subsequently, apply *Pseudomonas fluorescens* talc powder according to the plant age as described above, or spray *Pseudomonas fluorescens* culture solution to

drench the crown leaves.

Farmers use case:

When a farmer mentions using granules for the management of bud rot, it is important to ask what specific granules were used. If the product contains “*Chlorantraniliprole*”, “*Phorate*”, or “10G” or “3G”, please note that these are insecticides. However, bud rot is a fungal disease, not an insect-related issue.

- Kindly educate the farmer that using insecticides will not effectively control bud rot. Instead, recommend appropriate fungicidal control measures as per the standard guidelines for managing fungal infection.

Leaf Blight (Aku Endu Tegulu)

Symptoms:

- This disease appears as spots on older leaves and gradually spreads over the entire leaf. It predominantly affects the leaves in the outer whorl.
- Infected leaves start drying from the tip, and gradually the entire leaf appears dried out.
- Wave-like patterns appear on nuts infected with this disease, starting from the perianth end.
- This results in stunted, small nuts that eventually drop off. Yield loss ranges from 10 to 25 percent.
- This disease spreads throughout the year. Summer weather is particularly favorable for the disease. When weather conditions are conducive, fungal spores also spread through the air.

Management Practices:

- Remove infected leaves from the orchard and burn them.
- For coconut trees affected by this disease, apply 200 g of *Pseudomonas fluorescens* talc powder mixed with 50 kg of farmyard manure and 5 kg of neem cake annually.
- Spray a copper fungicide solution - 1% Bordeaux mixture or 0.1% Copper Oxychloride - twice at an interval of 45 days.
- Administer Hexaconazole (2 ml in 100 ml water) through root feeding once every 3 months.

- For controlling leaf blight on coconut seedlings less than 5 years old, mix 2 ml of Hexaconazole fungicide in 1 liter of water and spray, ensuring complete coverage of the leaf undersides.

Other Problems in Coconut Orchards - Control

1. **Button Shedding/Premature Nut Fall (Pinde Raludu):** Shedding of young nuts (buttons) is commonly observed, especially in young orchards that have just started bearing (8 to 10 years old). However, control measures should be taken when this shedding is excessive. Other causes like potassium deficiency, water stress, and hormonal imbalances also increase button shedding. Sometimes, pest infestations like scale insects, mealybugs, and eriophyid mites can also cause button shedding.
 - **Control:** To prevent button shedding, apply the recommended dose of potash fertilizer (2.5 kg/tree). Ensure the orchard does not suffer from drought by irrigating regularly. These practices can significantly reduce button shedding in coconut. If pests are identified, follow appropriate plant protection measures. Rats and squirrels also damage young nuts, causing them to fall. Placing rat poison like 'Roban' in the crown occasionally can control rats.
2. **Barren Nuts/Poorly Developed Nuts (Tattu Kayalu):** Occasionally, some trees in the orchard produce bunches where some nuts, or entire bunches, are barren ('ghost nuts')¹. These nuts fail to develop, remaining very small, or some nuts reach normal size but fail to form a copra inside². Farmers call these 'ghost nuts' or 'tattu kayalu'².
 - **Cause & Control:** These symptoms arise due to Boron or Potassium deficiency. Boron deficiency can also cause the crown leaves to fail to unfurl properly, remaining clumped together in the crown. For such trees only, apply 100 g of Borax powder per tree in the basin and irrigate. This can prevent the formation of barren nuts. Using the recommended dose of potash fertilizer (2.5 kg/tree) can increase nut size and copra yield.

Root feeding technique in coconut

Coconut root feeding is a valuable technique for delivering nutrients and pesticides directly to coconut, offering a targeted approach to address specific deficiencies and pest/disease issues. It's particularly beneficial for its efficiency, reduced chemical wastage, and minimal labor requirements compared to spraying.

Diseases and Pests Controlled by Root Feeding:

Root feeding is effective against a range of coconut pests and diseases, including:

Pests:

- Black-headed caterpillar (*Opisina arenosella*): This is a serious defoliating pest, and root feeding is a common control method.
- Red palm weevil (*Rhynchophorus ferrugineus*): While often managed with other methods like pheromone traps, root feeding can be effective, especially in early stages of infestation.
- Eriophyid mite (*Aceria guerreronis*): Root feeding with certain chemicals can help reduce mite populations.
- Slug caterpillar (*Parasa lepida*)
- Coconut skipper (*Gangara thyrasis*)
- Leaf caterpillars (*Turnaca acuta*)

Diseases:

- Basal Stem Rot (*Ganoderma lucidum*): Root feeding of fungicides in combination with soil drenching and neem cake application is effective.
- Stem Bleeding (*Ceratocystis paradoxa*): Root feeding of fungicides is recommended, especially when observed with *Ganoderma* infection.
- Root Wilt Disease (*Phytoplasma*): While management often involves removing severely infected coconut plants and replanting, root feeding of certain chemicals might be part of a broader strategy.

How to Perform Coconut Root Feeding:

1. Select a suitable root: Choose a fresh, live, fibrous root of pencil thickness. Dark brown colored roots are generally preferred.
2. Prepare the root: Make a sharp, slant cut at the end of the selected root. Some sources suggest a vertical cut to prevent rupturing the bag.
3. Prepare the solution: Mix the chosen chemical (pesticide or nutrient solution) with water in a polythene bag. The bag size should be appropriate (e.g., 15 x 4 cm for 40 cc solution).
4. Insert the root: Carefully insert the cut end of the root into the polythene bag containing the solution.
5. Secure the bag: Tie the mouth of the polythene bag tightly around the root using a cotton thread or soft wire to prevent spillage.
6. Position the bag: Bend the root downwards at an angle (e.g., 30°-40°) and secure it with a forked stick or by digging a deep enough hole to create space below the root. This helps in absorption and prevents spilling.
7. Monitor absorption: Allow the root to absorb the chemical for 24 to 48 hours. Check if the solution has been absorbed. If not, change the root and repeat the process.

8. Harvesting consideration: It's generally recommended to harvest mature nuts before administering the chemical. After root feeding, maintain a gap of at least 45 days before the next harvest of nuts to ensure chemical residues are below tolerance limits.

Important Note: Maintain a distance of 3.5 to 4 feet from the trunk when applying fertilizers or nutrients to the soil around the feeding roots. Dig a shallow basin, apply evenly, and water thoroughly.

Chemicals Used in Coconut Root Feeding (Latest Information):

The chemicals used for root feeding vary depending on the pest or disease being targeted. Always refer to the latest recommendations from agricultural universities or local extension services, as regulations and efficacy can change.

For Pests:

- Monocrotophos 36 SL: Traditionally used for black-headed caterpillars and red palm weevils. Dosage around 10 ml mixed with 10 ml water.
- Neem-based formulations (e.g., Azadirachtin 1% or 5%): Environmentally friendly option, often used for black-headed caterpillar and eriophyid mites. Recommended dosage might be 10 ml of 1% Azadirachtin + 10 ml water, or 7.5 ml of 5% Azadirachtin + 7.5 ml water. Can be applied thrice a year.
- Emamectin benzoate 5% SG: Effective against black-headed caterpillars.
- Bifenthrin 10% EC: Shows high efficacy against black-headed caterpillars.
- Imidacloprid 17.8% SL: Also effective against black-headed caterpillars.
- Profenofos 50% EC: Used for red palm weevil (e.g., 10 ml + 10 ml water).
- Carbosulfan 25 EC or Triazophos 40 EC: Used against black-headed caterpillars, typically 15 ml mixed with 15 ml water. Also, Triazophos (5 ml/litre) or Carbosulfan (2 ml/litre) can be sprayed for eriophyid mites, but root feeding is also an option for mites with Azadirachtin.
- Fenpyroximate 0.5% EC: Recommended for eriophyid mites (0.50 gm/tree through root feeding).

For Diseases:

- Fungicides for Basal Stem Rot/Stem Bleeding:
 - Aureofungin-sol (1.3g) + Copper sulphate (0.5g) in 100 ml of water: Often used for basal stem rot.
 - Tridemorph (2 ml in 100 ml water): Effective against basal stem rot and stem bleeding. Latest research suggests 2% Tridemorph via root feeding in combination with soil drenching.

- Thiophanate-methyl or Carbendazim (2.0 g in 100 ml of water): Can reduce basal stem rot incidence.
- Hexaconazole 5% + Validamycin 2.5% SC: Recent research shows this combination at 4 ml in 100 ml water for root feeding is effective against stem bleeding, especially when combined with soil drenching.
- Azoxystrobin 11% + Tebuconazole 18.3% SC: Another fungicide combination showing promise for stem bleeding (4 ml in 100 ml water).

For Nutrient Management:

- TNAU Coconut Tonic: A specific liquid nutrient mixture for coconut, recommended at 200 ml/palm once in six months for nut-bearing coconut plants. It helps improve chlorophyll content, photosynthetic efficiency, and controls button shedding.
- Potassium Chloride (KCl): Root feeding of 200ml of 1% KCl per tree thrice a year can help address potassium deficiency.
- Magnesium Sulfate (MgSO₄): Root feeding of 200 ml of 0.2% MgSO₄ twice a year can address magnesium deficiency.
- General Fertilizers: While often applied to the soil, the principle of root feeding can also be applied to nutrient solutions for targeted delivery of macro and micronutrients like nitrogen, phosphorus, potassium, magnesium, zinc, boron, and manganese.

Latest Information/Considerations:

- Integrated Pest Management (IPM): Root feeding should be part of a broader IPM strategy, which includes cultural practices (field sanitation, proper nutrition), biological control (using natural enemies), and mechanical methods (trapping, removal of infected parts).
- Residue Analysis: Research on chemical residues in coconut water and copra after root feeding is crucial to ensure food safety. Studies have shown that residues of Tridemorph applied via root feeding were lower than tolerance limits.
- Resistance Management: Relying on a single chemical for root feeding can lead to pest or pathogen resistance. Rotating chemicals or using combinations with different modes of action is advisable.
- Biopesticides: The use of biopesticides like Azadirachtin is gaining traction for root feeding due to their eco-friendly nature and effectiveness against certain pests.
- Precision Agriculture: Future research may focus on more precise methods of root feeding, possibly incorporating sensor technologies to monitor nutrient absorption and pest activity.

Always consult with local agricultural experts or extension services for specific recommendations tailored to your region and the particular issues affecting your coconut.

Trunk Injection

Yes, trunk injection (also known as endotherapy or vascular injection) is a recognized and effective method for treating certain pest and disease problems in coconut trees. This method involves directly delivering pesticides, fungicides, or nutrients into the tree's vascular system, allowing the compounds to be translocated throughout the plant.

Advantages of Injection Treatment:

- **Targeted Delivery:** The chemical is delivered directly into the tree, minimizing environmental contamination and exposure to non-target organisms (humans, wildlife, beneficial insects).
- **Reduced Drift:** Unlike spraying, there is no spray drift, making it safer for applicators and surrounding areas.
- **Effectiveness:** The systemic nature of the treatment ensures the chemical reaches the target pest or disease throughout the tree, even in tall coconut plants where spraying is difficult.
- **Long-lasting:** In many cases, injected treatments provide longer-lasting protection compared to external applications.

Commonly Used Chemicals and Doses:

The specific chemicals and doses depend on the pest or disease being targeted. Here are some examples:

For Pests (e.g., Coconut Rhinoceros Beetle, Red Palm Weevil, Black-headed caterpillar):

- **Monocrotophos (60% EC):** This is a systemic insecticide commonly used for trunk injection.
 - **Dose:** Typically, 10 to 20 ml of undiluted Monocrotophos (60% EC) per palm. Some recommendations suggest 10 ml mixed with 500 ml of water for pouring into the spindle of young coconut plants, or 20 ml mixed with an equal amount of water for root feeding. For trunk injection, 10 ml undiluted is often mentioned.
 - **Note:** Monocrotophos is highly toxic and its use may be restricted or banned in some regions due to its toxicity. It should be handled with extreme care and only by trained personnel.
- **Imidacloprid (e.g., IMA-jet, Imidacloprid 17.8% SL):** Another systemic insecticide effective against various sucking pests and beetles.

- **Dose:** Specific doses vary by product and tree size. For example, IMA-jet or IMA-jet 10 can be used for season-long preventative and curative control of Coconut Rhinoceros Beetle.
- **Emamectin Benzoate:** Effective against various pests, including CRB.
 - **Dose:** Varies by product. Can offer protection for up to a year or more.
- **Cartap Hydrochloride (50% SP):** This insecticide has shown effectiveness against the black-headed caterpillar.
 - **Dose:** All dilutions of Cartap Hydrochloride 50% SP injected into the base of the coconut palm effectively achieved 100% larval mortality for up to 90 days.

For Diseases (e.g., Bud Rot, Basal Stem Rot):

- **Fungicides (e.g., Fosetyl-Al, Potassium Phosphite, Copper oxychloride, Mancozeb):**
 - **Fosetyl-Al:** Effective against bud rot caused by *Phytophthora* species.
 - **Dose:** Injections of Fosetyl-Al have been effective in controlling bud rot, with annual injections being the norm, though intervals can be up to three years.
 - **Potassium Phosphite:** Also effective against *Phytophthora* species and can stimulate plant defense mechanisms.
 - **Dose:** Studies have used 30 mL (8.4 g a.i.), 60 mL (16.8 g a.i.), and 90 mL (25.2 g a.i.) of potassium phosphite diluted 1:2 with deionized water.
 - **Aureofunginsol + Copper Sulphate:** For Basal Stem Rot (*Ganoderma lucidum*).
 - **Dose:** 2g Aureofunginsol + 1g Copper Sulphate in 100 ml of water. This is usually applied through stem injection or root feeding quarterly for one year.
 - **Tridemorph:** Also for Basal Stem Rot.
 - **Dose:** 2ml in 100ml of water, applied quarterly for one year through stem injection or root feeding.

Precautions to be Taken:

1. **Read the Label Carefully:** Always follow the specific instructions, dosage rates, and safety precautions provided on the pesticide/fungicide label. This is paramount for effectiveness and safety.
2. **Personal Protective Equipment (PPE):** Wear appropriate PPE, including gloves, eye protection, long-sleeved shirts, and pants, to prevent skin and eye exposure. Respirators might be necessary for certain chemicals.
3. **Timing of Application:**

- For pests, treatment is most effective during the early stages of infestation.
 - For diseases like bud rot, prophylactic sprays or injections before the onset of monsoon are often recommended.
4. **Harvesting Nuts:** If the infested palm bears nuts, harvest all mature nuts before insecticide application. Some insecticides have a waiting period (e.g., 45 days) before nuts can be harvested for consumption. Always check the label.
 5. **Drilling and Injection Technique:**
 - Use a clean, sharp drill bit (e.g., 13 mm diameter) to make holes at a 45° angle, about 1 meter above ground level and 10 cm deep. Make two holes on opposite sides of the trunk for systemic distribution.
 - Ensure the holes are sealed after injection to prevent leakage and promote healing. Dowels or appropriate sealants can be used.
 - Sterilize tools with diluted bleach before and after use to prevent the transfer of disease pathogens.
 6. **Avoid Overdosing:** Using higher than recommended doses can harm the tree and lead to unnecessary chemical residues.
 7. **Environmental Considerations:**
 - While injection reduces environmental exposure compared to spraying, dispose of chemical containers properly.
 - Be aware of any local regulations or bans on specific chemicals.
 8. **Professional Guidance:** For severe infestations or unfamiliar situations, consult with agricultural experts or plant protection specialists. They can provide accurate diagnosis and recommend the most suitable treatment plan.
 9. **Tree Health:** Trunk injection causes a permanent wound. This method is generally recommended for prevention of lethal diseases or control of significant pest outbreaks. Consider the overall health and vigor of the tree before applying injections.

By adhering to these guidelines, injection treatment can be a safe and highly effective method for managing pests and diseases in coconut trees.

where will i get quality coconut plant material in andhra pradesh:

- Ambajipeta (East Godavari District): Horticultural Research Station, Dr. Y.S.R. Horticultural University, Ambajipeta - 533214.
- East Godavari District Horticulture Officer: B. Sujatha Kumari, Assistant Director of Horticulture. Mobile: 7995086768. Email: dhorjvm@gmail.com
- Baruva (Srikakulam District): Kobbari Naru Kendram, Opp. Kanakadurga Temple & Marine Police station, Baruva Beach Road, Baruva (V), Sompeta (M). Pincode: 532264.
- Contact Person: Sri Shankar Das, Horticulture Officer. Mobile: 7995086806

- Bobbili (Vizianagaram District): Near TCY ITI Colony, Beside Govt. ITI Bobbili (V & M), Vizianagaram District. Pincode: 535558.
- Contact Person: Smt. Venkata Ratnam, Horticulture Officer. Mobile: 7995086817, 9490577361
- Vizianagaram District Horticulture Officer: A.V.S.V. Jamadagni, Deputy Director of Horticulture. Mobile: 7995086762. Email: ddhvzm@yahoo.com
- Anakapalli (Anakapalli District): Kobbari Naru Kendram, Opp. Union Bank of India, Gundala street, Anakapalli (V & M), Anakapalli District. Pincode: 531001.

Contact Person: Smt. Lavanya, Horticulture Officer. Mobile: 7995086826

Anakapalli District Horticulture Officer: G. Prabhakara Rao, Assistant Director of Horticulture. Mobile: 7995086764. Email: dhoakp22@gmail.com

- Tetagunta (Kakinada District): Horticulture farm, Tetagunta (V), Tuni (M), Kakinada District. Pincode: 533406.
 - Contact Person: Sri Aruna, Horticulture Officer. Mobile: 7995086854
 - Kakinada District Horticulture Officer: N. Mallikarjuna Rao, Assistant Director of Horticulture. Mobile: 7995086765. Email: dhokakinada@gmail.com
- Vellaturu (NTR District): Model Nursery, Vellaturu (V), G Konduru (M), NTR District. Pincode: 522257.
 - Contact Person: Sri Parameshwar Reddy, Horticulture Officer. Mobile: 7995086880
 - NTR District Horticulture Officer: P. Balaji Kumar, Assistant Director of Horticulture. Mobile: 7995086772. Email: adh_vijayawada@yahoo.co.in
- Darsi (Prakasam District): Horticulture Farm, Darsi Kurichedu road, Sivaji Nagar (V), Darsi (M), Prakasam District. Pincode: 523247.
 - Contact Person: Sri Ravi Venkanna Babu, Horticulture Officer. Mobile: 7995086922
- Kothapatnam (Prakasam District): Horticulture Farm, Kothapatnam, Eethamukkala road, Opp HP Gas company, Near Power Station Kothapatnam (V), Kothapatnam (M), Prakasam District. Pincode: 523286.
 - Contact Person: Sri Teja, Horticulture Officer. Mobile: 7995086929
- Gudipallipadu (SPSR Nellore District): Horticulture Farm, Gudipallipadu, Narayana reddy Pet, Narukuru cross road, Opp. Sai Marine company Nellore Rural, SPSR Nellore District. Pincode: 524314.

- Contact Person: Prasanna, Horticulture Officer. Mobile: 7995086942
- SPSR Nellore District Horticulture Officer: M.V.Subba Reddy. Mobile: 7995086780. Email: dhonellore@gmail.com
- Srikalahasthi (Tirupati District): Horticulture Farm, Srikalahasthi, Beside RTO Office, Cherlopalli (V), Srikalahasthi (M), Tirupati Dist. Pincode: 517641.
 - Contact Person: Smt. Pushpalatha, Horticulture Officer. Mobile: 7995086968

General Contact for Andhra Pradesh Horticulture Department:

- Commissioner of Horticulture, Andhra Pradesh:
 - Email: horticulturedept@yahoo.co.in
 - Office Phone: 0863-2216470
 - Main Contact Page for AP Horticulture Department:
<http://horticulture.ap.nic.in/contact.html> (This page lists various officers and their contact numbers, which can be useful for general inquiries or to be directed to the right person for specific nursery information).

Private nurseries which are listed popular in the search results:

- Sri Lakshmi Nursery Gardens: East Godavari/Rajahmundry. Contact: +91-8048951367 (general inquiry, confirm directly).
- Greens Ecom & Nursery Exports Pvt. Ltd.: Kadiyam, East Godavari. Contact: +91-8048951367.
- SATYADEVA NURSERY - PULLA CHANTIYYA: Kadiyapulanka, East Godavari. Search Justdial for contact.
- Mahindra Nursery (Kadiyam Nursery & Plantation): Kadiyam, Rajahmundry. Contact: +91-9493616161. Website: mahindranursery.com

Check variety, plant quality, pricing, minimum order, and transport. Visiting the nursery is recommended.

Whom to Contact for coconut schemes, Government Support for New Coconut Farmers Setting up a Coconut Nursery

The primary authority to contact for coconut cultivation in India is the **Coconut Development Board (CDB)**. They are a statutory body established by the Government of India under the Ministry of Agriculture & Farmers Welfare, dedicated to the integrated development of coconut cultivation and industry in the country.

You can reach out to them through:

- **Regional Offices and State Centres:** The CDB has regional offices and state centers across various states, including Tamil Nadu, Bihar, Maharashtra, and more. You can find their contact details on the official CDB website.

The Coconut Development Board has a State Centre in Andhra Pradesh:

Address: D.No.54-14/5-18A, Road No.11, Bharati Nagar, Near Novotel Varun Hotel, Ring Road, NTR District, Vijayawada 520008, Andhra Pradesh

Phone: (0866) 2472723

Email: sc-vijayawada(at)coconutboard(dot)gov(dot)in

You can find more information on the [Coconut Development Board website](#).

State Horticulture Department: Your state's Horticulture Department also plays a crucial role. They often implement CDB schemes at the state level and can provide local guidance and assistance. For example, in Andhra Pradesh, you can contact the District Horticulture Officer or Mandal/Divisional Level Horticulture Officers in your area. You can find contact information for specific districts on your state's horticulture department website (e.g., horti.ap.nic.in for Andhra Pradesh). (list of Department staff to be uploaded)

- **"Hello Naariyal" Call Center:** The CDB has launched a call center called "Hello Naariyal" to cater to the needs of coconut farmers regarding harvesting and other plant management practices. You can contact them at **0484-2377266 (Extn: 137)**.

The Indian government, primarily through the Coconut Development Board and State Horticulture Departments, offers various schemes and financial assistance to encourage coconut cultivation and support new farmers. Here are some key areas of support:

- **Production and Distribution of Quality Planting Material:**
 - **Subsidies for Seedlings:** Schemes provide assistance for procuring quality coconut saplings from government or CDB-assisted nurseries. For instance, some schemes offer a percentage of the cost per seedling.
 - **Establishment of Nurseries/Seed Gardens:** Financial assistance is available for individual farmers, cooperative societies, and other organizations to establish small or nucleus coconut seed gardens and nurseries for producing quality planting materials.
- **Expansion of Area under Coconut:**
 - **New Planting Subsidy:** You can receive subsidies per hectare for bringing new areas under coconut cultivation. This assistance is often provided in installments.

For example, some schemes offer up to 25% of the cost for a maximum of 10 acres per beneficiary.

- **Integrated Farming for Productivity Improvement:**
 - **Demonstration Plots:** Financial assistance is provided for laying out demonstration plots to showcase improved cultivation practices.
 - **Organic Manure Units:** Support may be available for constructing organic manure units to promote sustainable farming.
- **Replanting and Rejuvenation of Coconut Gardens:**
 - **Financial Assistance:** Schemes provide support for cutting and removing old or unproductive coconut plants, replanting with new seedlings, and implementing integrated management practices for existing gardens. This can include assistance for integrated nutrient management (INM) and integrated pest management (IPM).
- **Coconut Palm Insurance Scheme (CPIS):**
 - This scheme helps farmers insure their healthy, nut-bearing coconut against natural perils leading to death or unproductivity. The Board typically bears 50% of the premium, with the remaining shared between the State Government and farmers.
- **Minimum Support Price (MSP) for Copra:**
 - The government fixes a Minimum Support Price for copra (dried coconut meat) to ensure remunerative prices for coconut growers and incentivize production.
- **Training and Technical Guidance:**
 - The CDB and other agricultural departments offer training programs and technical guidance on scientific coconut cultivation, disease management, value addition, and market development.

To avail these schemes, you typically need to contact your local Horticulture Officer or the Coconut Development Board. They will guide you through the application process, required documents (like land records, Aadhaar card, bank passbook), and eligibility criteria. Many states also have online portals (like I-Khedut in Gujarat) where you can apply for agricultural schemes.

Burnt appearance in orchards :

Based on your description, the pest you're likely dealing with in your coconut plantation is the **Coconut Leaf Caterpillar**.

Here's why:

- **Dried spots on leaves leading to complete drying:** The larvae of the coconut leaf caterpillar (especially *Opisina arenosella*, also known as the black-headed caterpillar) scrape the green matter from the leaves, leaving behind dried, brown patches that eventually cause the entire leaf to wither and look "burnt."

- **Entire plantation looking burnt:** A severe infestation can make the whole plantation appear scorched due to widespread leaf damage.
- **Caterpillars scraping and eating leaves from underneath:** This is a classic symptom of the coconut leaf caterpillar. The caterpillars feed from the underside of the leaflets, often within silken galleries or tunnels they construct.

This pest can cause significant damage if not managed effectively.

The predominant acid found in coconut, specifically in coconut oil, is **Lauric acid**.

coconut in India are susceptible to several destructive diseases, among which “Thanjavur Wilt” and “Tatipaka” disease are significant concerns. Here's a detailed look at both, with the latest available information:

“Thanjavur Wilt ” (Basal Stem Rot / Ganoderma Wilt / Anabe Roga)

Latest Information:

- **Ongoing Threat:** Thanjavur wilt remains a major limiting factor in coconut production, particularly in the coastal districts of Tamil Nadu, with incidence rates ranging from 6.5% to 50% in districts like Thanjavur, Nagapattinam, and Thiruvarur. (International Journal of Plant & Soil Science, 2023)
- **Post-Cyclone Spread:** This disease has shown a severe spread in the East Coast region of Tamil Nadu after cyclones, highlighting the need for continued research and management efforts. (International Journal of Plant & Soil Science, 2023)
- **Research Focus:** Decades of research have been conducted on this disease, focusing on isolation of the pathogen, pathogenicity, virulence, disease index development, early detection, epidemiology, pathophysiology, and various management strategies. (International Journal of Plant & Soil Science, 2023)

Causal Organism: Thanjavur Wilt is primarily caused by the soil-borne fungi *Ganoderma lucidum* and *Ganoderma applanatum*.

Symptoms: The disease progresses through distinct stages, often leading to the death of the palm within 6 to 54 months (average 24 months) from the appearance of bleeding patches.

- **Initial Symptoms (Stage I):**
 - Wilting and yellowing of the outermost whorl of leaves, which may not always be prominent.
 - Decay and death of fine roots.
- **Intermediate Stages (Stage II & III):**

- **Bleeding Patches:** Characteristic reddish-brown liquid exudes from cracks at the base of the stem, near the ground level, gradually extending upwards. The tissues beneath these patches are soft and decaying.
- Extensive root decay and discoloration (roots become watery with an alcoholic smell, red below the hypodermis, and brownish towards the stele).
- Reduction in the production of new bunches and heavy button shedding.
- Drooping of lower leaf whorls.
- Nuts become barren.
- **Advanced Stages (Stage IV & V):**
 - Stem decay extends upwards, and the lowest leaf whorls dry and fall off.
 - Other leaves also droop, leaving only the spindle leaf and a few young leaves erect.
 - Eventually, all leaves droop and fall, leaving a shriveled, dried, decapitated stem.
 - Sporophores (bracket formations) of *Ganoderma* may appear at the base of the trunk.
 - Scolytid beetles (*Xyleborus perforans*) and weevils (*Diocalandra stigmaticollis*) are often found boring into the stem at bleeding patches in later stages, accelerating palm demise.

Spread:

- Mainly through root contact between diseased and healthy coconut plants.
- Favored by repeated ploughing and uncontrolled flood irrigation.
- More common in sandy or sandy loam soils, neglected plantations, and in areas with old infections.

Management: Integrated disease management is crucial for effective control:

- **Cultural Methods:**
 - **Removal and Destruction:** Promptly remove and burn severely infected coconut plants to prevent further spread.
 - **Isolation Trenches:** Dig a trench (50 cm wide and 1 m deep, 2 to 3 m away from the base of the trunk) around diseased coconut plants to prevent root contact and fungal spread.
 - **Irrigation Management:** Adopt ring irrigation to avoid water flow from diseased to healthy coconut plants. Avoid flood irrigation. Irrigate adequately during summer months.
 - **Green Manuring/Organic Amendments:** Grow green manure crops (e.g., cowpea, sunhemp) and incorporate them *in situ* at flowering. Apply ample

farmyard manure (FYM) or compost (50 kg/palm/year) and neem cake (5 kg/palm/year). These enhance soil health and support beneficial microorganisms.

- **Avoid Mechanical Injury:** Prevent injury to the trunk during cultural operations.

- **Biological Methods:**

- Apply antagonistic microorganisms like *Pseudomonas fluorescens* (200 g/palm/year) and *Trichoderma viride* (200 g/palm/year) mixed with FYM.
- Apply Phosphobacteria and Azotobacter (200 g each mixed with 50 kg FYM/palm/year) to enhance soil fertility and palm resistance.

- **Chemical Methods:**

- **Root Feeding/Trunk Injection:** For early stages, root feeding with Tridemorph 20% EC (10 ml+15 ml water/tree) or Aureofunginsol 21% WP (1 g+1 g Copper Sulphate in 100 ml water/tree) or Calixin (3 ml/tree) at quarterly intervals for one year.
- **Soil Drenching:** Drench the soil around the trunk (1.5 m radius) with 40 litres of 1% Bordeaux mixture quarterly, repeated after 2 to 3 years.
- **Chiseling and Protection:** Knife out bleeding patches, burn the exposed tissue, apply Tridemorph (5% Calixin), and then protect with hot coal tar or Bordeaux paste.

“Tatipaka” Disease

Latest Information:

- **Phytoplasmal Etiology:** Tatipaka disease is confirmed to be caused by **phytoplasmas**. Transmission Electron Microscopy has revealed the presence of pleomorphic, triple-layered membrane particles (ranging from 100 nm to 1000 nm) in the phloem vessels of diseased leaf samples.
- **Irregular Distribution:** Phytoplasmas were detected in leaf samples but not consistently in peduncle, spathe, or nut samples, suggesting an irregular distribution within the diseased palm's tissues. Further research on their distribution and transmission is needed.
- **Low Incidence:** Recent surveys indicated that Tatipaka diseased coconut plants were at a negligible level (below 1%), and eradication efforts for identified infected coconut plants are being undertaken to prevent further spread.
- **Mid Whorl Yellowing Disease (MWYD):** It's important to note that a separate disease, "Mid Whorl Yellowing Disease" (MWYD), prevalent in southern Kerala, exhibits some similar symptoms (yellowing, abnormal nut fall) but has been identified as distinct from root wilt phytoplasma. Its etiology is still under elaborate study. (Journal of Advances in Biology & Biotechnology, 2025)

Causal Organism: Phytoplasmas (formerly referred to as Mycoplasma-like organisms or MLOs). These are non-helical, non-culturable plant pathogenic bacteria transmitted by insect vectors (arthropods).

Symptoms: Tatipaka disease primarily affects coconut plants in the 20 to 60 year age group; young coconut plants are rarely affected.

- **Underground Symptoms:** Reduction in the number and size of roots, and extensive root rot.
- **Above-ground Symptoms:**
 - Tapering of the stem below the spindle.
 - Reduction in the size and number of leaves.
 - Reduction in the number of bunches and overall spindle size.
 - Abnormal shedding of buttons and immature nuts.
 - The nuts from diseased coconut plants have thinner husks, weaker fibers, and the copra remains soft and flexible instead of drying into hard copra.
 - Tender coconut water from affected coconut plants can be insipid.

Spread:

- It is believed to be spread by insect vectors, though specific vectors for Tatipaka are not always explicitly mentioned in general coconut disease management. Phytoplasmas are typically vector-borne.
- The disease was first noticed in Tatipaka Village in East Godavari District, Andhra Pradesh, after a cyclone in 1949.

Management: Due to the phytoplasma nature and often low incidence, management strategies for Tatipaka disease often overlap with general phytoplasma disease management in coconuts, which can include:

- **Eradication:** Identifying and removing infected coconut plants to prevent further spread, especially if the incidence is low.
- **Vector Control:** Managing insect vectors that transmit phytoplasmas. Specific insecticide treatments may be used in the leaf axils.
- **Cultural Practices:**
 - Maintaining good garden hygiene.
 - Ensuring adequate nutrition and irrigation.
 - Growing green manure crops in basins.
- **Antibiotics (Temporary Relief):** While antibiotics can cause temporary remission, the disease often reappears after treatment cessation, making them not a sustainable long-term solution.

- **Breeding for Resistance:** Long-term strategies often involve developing disease-resistant varieties.

It's important for farmers to regularly inspect their coconut for any signs of disease and implement integrated management strategies as advised by local agricultural extension services.

Grey blight in coconut is primarily caused by the fungus **Pestalotiopsis spp.** (specifically, *Pestalotiopsis palmarum*).

This fungal disease results in characteristic greyish spots with dark brown margins on the coconut leaves, and in severe cases, can lead to widespread leaf blight.

11. Additional Problems

- **Tethering Cattle:** Provides weed control and fertilization but risks root damage and pest breeding (e.g., rhinoceros beetle grubs in manure piles).
- **Fish Ponds:** Increase humidity, potentially promoting pests Black headed caterpillars, red palm weevil, mealybugs, and mites.

Rats& Rodents To effectively control rats and rodents in coconut plantations in Andhra Pradesh, farmers can adopt a practical, integrated approach to minimize crop damage from pests like the Indian field rat and bandicoot rat, which gnaw on tender nuts and cause significant yield losses. Wrapping smooth iron or galvanized metal sheets (30 to 40 cm wide) around tree trunks at a height of 1.5 to 2 meters prevents rats from climbing to reach nuts, as they cannot grip the surface. Maintain plantation hygiene by clearing weeds, fallen nuts, and debris, and avoid unmanaged cattle manure piles, which attract rats; turn manure heaps every three months and cover with nylon nets. Set up snap or glue traps baited with coconut pieces or peanut butter near tree bases or rat runways, checking and resetting them regularly. For severe infestations, use anticoagulant rodenticides like bromadiolone (0.005%, 10 to 20g per bait station) in secure stations, ensuring they are inaccessible to children, pets, and wildlife, and dispose of dead rats promptly to prevent secondary poisoning. Encourage natural predators like owls by installing nest boxes, and grow intercrops like turmeric, ginger, or green manures (e.g., sunn hemp) to reduce open spaces where rats thrive. Avoid damaging coconut roots during weeding or trap placement to prevent disease susceptibility. Farmers can seek guidance from the Horticultural Research Station in Ambajipeta (533214) or the Coconut Development Board in Vijayawada (0866-2472723) for tailored advice and resources. Regular monitoring for gnaw marks and droppings ensures timely adjustments to these measures, protecting coconut yields effectively.